

One Earth Village Wellbeing & Balance Sheet: Comprehensive Insights Strategy

Executive Summary

The One Earth Village Wellbeing project is fundamentally shifting how community development is measured, managed, and accelerated. By moving away from fragmented, retrospective surveys towards a unified **Village Balance Sheet** and **Wellbeing Scoring System**, stakeholders can now see a real-time, holistic picture of a community's true wealth—spanning financial, natural, social, and human capital.

This document synthesizes the strategic insights we derive from this integrated framework, details the critical data points required to power it, and explains exactly how these data points translate into actionable wellbeing scores and balance sheet liabilities/assets.

Strategy Layer Purpose

The core purpose of the strategy layer is to assess where households and villages are in their wellbeing journey and to empower them to progress meaningfully.

All outputs from this framework should be framed as decision-support insights for evidence-based planning and action.

The system evaluates:

- What people currently have (assets, capabilities, access)
- What they need (gaps, vulnerabilities, constraints)
- What potentials are possible (latent capacities, regenerative opportunities)

1. Core Strategic Insights Derived from the Project

By deploying this integrated assessment framework, stakeholders (from village leaders and NGOs to government bodies and technical teams) unlock several transformative insights:

A. The "True Wealth" Insight (Beyond GDP)

Traditional economic metrics fail to capture the true health of a village. Our framework provides the insight that a village might be financially poor but ecologically and socially wealthy.

- **The Insight:** We can quantify the exact value of intact forest cover, strong community support systems, and local food self-sufficiency, ensuring these assets are recognized and protected during economic development.

B. The Minimum Material Needs Liability

Through our newly developed **Minimum Material Needs Income Indicator**, we calculate the absolute baseline expenditure required for a household to survive (water, energy, basic food, housing upkeep).

- **The Insight:** Rather than just tracking income, we track the *gap*. This gap is treated as a strict **Liability on the Village Balance Sheet**. Stakeholders instantly know exactly how much capital injection or structural subsidy is required to lift the village above the survival threshold.

C. The Capability and Growth Bottleneck Insight

By mapping "Aspirations," "Wishes," and "Unfulfilled Needs" directly onto the balance sheet as liabilities, we identify the exact friction points preventing a community from thriving.

- **The Insight:** If a village has high agricultural assets but severe unfulfilled educational needs, the insight is that capital should not be poured into more farming equipment, but rather into intellectual and developmental infrastructure to unlock the next stage of growth.

D. Systemic Multi-Capital Resilience

By tying environmental scores (water harvesting, renewable energy) directly to economic durability, we map systemic resilience.

- **The Insight:** We can predict how a village will weather an economic or climate shock based purely on their Natural and Human Infrastructure balance sheets, allowing for highly targeted preventative interventions.

2. Critical Data Points Required

To generate these insights, the system relies on a meticulously structured Data Architecture Map containing over 400 tracked parameters across the household and village-head levels.

Priority 1: Critical Balance Sheet & Wellbeing Data

These data points form the absolute foundation of the computations.

- **Demographics & Material Wealth:** Total population and household size (Q0203), age distribution, and exact housing footprints/space per capita (Q0294).
- **Financial Flows:** Annual household income (Q0212), diversified income streams (Q0034, Q0048, Q0068, Q0092, Q0105, Q0118 for farm; Q0411 for business), and baseline expenditures (Q0316, Q0320, Q0324, Q0328 for water; Q0340 for energy).
- **Natural Resource Inventories:** Land area owned versus utilized (Q0266-Q0288), forest cover canopy estimates (Q0362), daily water access quality (Q0315), and timber harvest rates (Q0363).
- **Energy Infrastructure:** Grid connectivity limits (Q0338), micro-grid installations (Q0348), and reliance on alternative organic/wood fuels (Q0347, Q0349).
- **Food Production vs. Consumption:** The delta between crop/livestock yield (Q0020-Q0130) and the dietary diversity / caloric intake of the village (Q0131-Q0202).

Priority 2: Liabilities & Contextual Data

These points determine the "needs" side of the ledger.

- **Unfulfilled Needs & Aspirations:** Self-reported gaps in education, healthcare, infrastructure, and economic opportunity, tracked directly as balance sheet liabilities (Q0243-Q0251 for Aspirations; Q0252-Q0259 for Unfulfilled Needs; Q0260-Q0265 for Wishes).
- **Mobility & Accessibility:** Vehicle ownership vs. transportation needs (Q0351-Q0361).
- **Community Infrastructure:** Distance and access to primary schools, clinics, banks, and waste management systems (External / Village Head macro data).

Priority 3: The Intangible Wellbeing Qualities

These points finalize the holistic wellbeing score.

- **Social & Emotional Metrics:** Frequency of community gatherings, strength of individual support systems, self-reported stress levels, and empathy markers (Q0218-Q0223).
- **Intellectual & Spiritual markers:** Traditional knowledge retention, active skill-seeking behavior, and community harmony scores (Q0224-Q0242 for Intellectual; Q0251/Q0221 for Spiritual/Resilience).

Structured Data Sources That Power the Model

The framework should draw from a structured mix of sources so that scores and insights are both grounded and extensible:

- **Survey-based primary data:** Household and village responses captured through the app and field tools.
- **Community-validated participatory data:** Locally confirmed records, village-head inputs, and participatory review.
- **Government registries and administrative datasets:** Household, land, occupation, health, education, and infrastructure records where available.
- **Standardized global registries and code lists:** Shared dropdown standards and reference taxonomies for interoperability.
- **Remote sensing and GIS layers:** Satellite imagery, land-cover maps, flood-risk zones, and topographic data.
- **IoT and field equipment:** Water testing kits, soil sensors, meters, and climate or air-quality instruments.
- **Threshold and benchmark references:** The project's threshold layer already draws on WHO, UN-Habitat, FAO, IUCN, SDG frameworks, World Bank references, and Malaysian standards through `thresholds/village_level_thresholds.md`.

3. The Bridge: How Data Connects to Wellbeing & Balance Sheets

The brilliance of the One Earth framework lies in its translation engine. Raw data points are never just stored; they are transformed into actionable ledgers and scores.

3.1 Translating Data to the Balance Sheet

The **Village Balance Sheet** tracks physical, natural, and financial capital using a Surplus/Deficit logic.

- **Assets:** Owned land, local businesses, renewable energy infrastructure, intact forests, and local food yields.
- **Liabilities:** The *Minimum Material Needs Income Indicator* (the cost of required expenditures), housing deficits, unmet water demands, reliance on imported energy, and the quantified "Unfulfilled Needs" (e.g., the cost of building a requested clinic).
- **The Computation:** The AI engine compares Assets vs. Liabilities to generate a **Surplus Ratio**. A village that produces 120% of its required food but only 40% of its required energy shows a clear, imbalanced ledger.

3.2 Translating Data to the Wellbeing Score

The **Wellbeing Scoring System** measures the lived experience of the villagers across 7 Spheres (Physical, Occupational, Emotional, Social, Environmental, Intellectual, Spiritual).

- **The Computation:** Using defined 0-1 formulas, raw data is normalized. For instance, an income below the *Minimum Material Needs* threshold automatically restricts the Occupational Wellbeing score to a maximum of 0.3. Conversely, high community participation and shared harvests immediately boost the Social Wellbeing score to 1.0.
- **The Blending:** The individual wellbeing scores of the residents are aggregated and blended with the physical Village Balance Sheet Resource Score (at a 60/40 ratio) to produce the ultimate **Village Sustainability Score**.

3.3 The Final Loop: AI Advisory and Local Empowerment

Ultimately, this data flows back to the community.

- **Present Day:** Village heads receive actionable dashboards showing exactly where their balance sheet leaks are occurring.
- **The Future:** As the dataset grows, the integrated AI engine will serve as a continuous advisory platform, automatically connecting villages with identified balance-sheet deficits to appropriate marketplaces, government grants, and external knowledge resources.

Conclusion: Every data point requested from the community serves a mathematical purpose. By tracking both tangible assets and intangible aspirations on a unified ledger, the One Earth system guarantees that financial development and holistic human wellbeing progress together, ensuring no community is left behind.